Fiscal Year Project Funded: 2015

Project Name:
"Science-based Management of Invasive Mountain Goats on Kodiak Island"

Project Location:
Kodiak National Wildlife Refuge, Kodiak Island, Alaska

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Project Goal:
The invasive mountain goat population on Kodiak Island continues to grow exponentially and has now reached a critical threshold. If the population exceeds carrying capacity, it will over-utilize native forage resources, crash to a new lower carrying capacity, and likely impact fragile native flora and fauna (i.e. Caughley's irruptive growth pattern). These outcomes are highly inconsistent with the management goals of the Kodiak National Wildlife Refuge and the Alaska Department of Fish and Game (ADF&G). To avoid this outcome, and develop an empirically-based harvest management plan, managers need a detailed understanding of mountain goat abundances, population dynamics, and resource selection patterns.

Expected Conservation Outcome of the Project:
An empirically-based harvest management plan for introduced mountain goat on Kodiak Island that limits population growth and avoids potentially harmful ecological impacts from an overabundant population.

Project Measureable Objectives (Year 1):
We fit 31 mountain goats with GPS collars in July 2015. We are monitoring collared animals directly via fixed-wing aerial surveys and indirectly via satellite transmissions.

Project Measureable Objectives (Long -term): We will use data from GPS collars to develop an aerial survey sightability model, and to quantify population dynamics and resource selection patterns for mountain goats on Kodiak Island. These results will be used to develop an empirical model for harvest management.

Assessment of Short-term Performance (Year 1):
o Less than 70% conservation objective achieved
o 70 – 79% conservation objective achieved
o 80 – 89% conservation objective achieved
X 90% or more conservation objective achieved

Assessment of Long-term Performance this year:
o Less than 70% conservation objective achieved
X 70 – 79% conservation objective achieved

o 80 – 89% conservation objective achieved
o 90% or more conservation objective achieved

Project Status:
We captured mountain goats from a Eurocopter AS350B A-Star helicopter (Soloy Helicopters, Palmer, AK). We immobilized goats with carfentanil citrate delivered via a 2-cc barbed dart fired from a projector (Palmer Cap-Chur Equipment, Powder Springs, GA). Prior to darting, we manipulated goat groups with the helicopter into terrain that enabled close-range darting when animals were stationary (i.e., hunkered up against a small terrain feature or pausing to look back). By limiting darting events to these periods, we increased our odds of ideal dart placement (i.e. hindquarter), which produces short induction times (<4 minutes) that maximize animal safety.

We fitted captured goats with either a) a single Globalstar GPS collar (W300 Wildlink-GTX, Advanced Telemetry Systems, Isanti, MN), or b) a spread-spectrum GPS collar (TGW-4590, Telonics, Mesa, AZ) and a VHF collar (MOD-400-1, Telonics, Mesa, AZ). We programmed Globalstar collars to collect two daily fixes, each separated by 12 hrs. Fixes cycled one hour forward daily (i.e., 12:00 on Day 1, 13:00 on Day 2) to reduce temporal bias associated with collecting fixes at the same time of day. We programmed spread-spectrum collars to record a fix every 3 hours between 16 December and 16 May, and every six hours between 16 May and 16 December. Six hours of collar inactivity triggered a mortality alert in all collars. Globalstar collars transmitted fix and mortality status datasets daily via a satellite to a website, where we access them. Alternatively, spread-spectrum collars required us to manually upload data to a laptop computer in a fixed-wing aircraft. We programmed release mechanisms (CR-2A) to release spread-spectrum collars from goats three years after capture (9 August, 2018). VHF collars did not have a release mechanism and had a predicted battery life of 3 additional years (expiring 2021).

We fitted goats with a large numbered red tag on each ear for identification purposes. We collected ear tissue samples for genetic analyses, and we collected blood samples for trace mineral, serology and other analyses. We collected nasal and throat swabs to test for infectious diseases. We collected fecal pellets to assess diet (microhistological analysis). We estimated rump fat thickness using a portable ultrasound device. We estimated age by counting horn annuli. We measured individual horn annuli thickness to assess annual horn growth. We measured body mass with a digital scale, aided by a collapsible weighing bar and a net bag. We measured body length and girth, and neck and hind foot length. We assessed female kidding success by checking lactation status. We delivered injected antibiotics (oxyetracyclin, 9 mL for females and 15 mL for males) intramuscularly at two sites in the rump.

To insure that captured goats were stable throughout the handling process, we monitored internal body temperature with a rectal thermometer and dissolved blood oxygen with a pulse oximeter on their tongue. If temperatures exceeded 105^{o}F, we expedited handling and attempted to cool the goat using snow, ice and/or water. To reverse immobilization drug effects after processing or when temperatures exceeded 107^{o}F, we hand-injected 255 mg Naltrexone (5.1 cc) intramuscularly in the rump. Reversal times vary, but goats generally stand within 3-4 minutes.

We monitored Globalstar collared goats every 3-4 weeks since capture by downloading fix locations and mortality status from the web. We plan on monitoring spread-spectrum collars on a 6-week interval (fixed-wing survey) throughout the winter to download collar data and determine live/dead status.

We explored goat space-use patterns by examining 95% minimum convex polygons (MCPs) and 95% fixed kernels home ranges.

Report Summary:
We captured 31 mountain goats (12 males, 19 females) from 20-26 July, 2015 (Table **1**). We fit 26 goats with ATS Globalstar collars and five goats with Telonics spread-spectrum collars. We immobilized all goats with 2.55 mg carfentanil citrate delivered using brown charges. Induction time averaged 3 min 30 sec, and ranged from 2 to 20 minutes. Goat were immobilized (time down) an average of 35 minutes. Captured goats averaged 6 years old, and ranged from 1 to 12 years. The average weight was 91 kg (200 lbs), and was 75 kg (165 lbs) for female and 116 kg (255 lbs) for male goats.

Weather conditions during the capture period ranged from clear and sunny with temperatures into the mid- to high 60^{o}F, to cloudy with light rain with temperatures in the high 50^{o}F. The Kodiak Archipelago had a lower than normal snowfall in 2015, so there was little snow cover below 600 m (2000 ft). Capture elevations averaged 653 m (2,144 ft), and ranged from 338 m (1,112 ft) to 908 m (2,980 ft).

By December 2015, 28 of 31 collared goats (90%) of collared goats were alive. All three mortalities (males) were legally harvested by hunters between 2 October and 11 November. This included one captured on the Aliulik Peninsula, one captured south of Karluk Lake, and one collared on Brown’s Ridge. We continue to monitor the remaining collars (n=28).

Mountain goat 95% MCP home ranges averaged 36 km^2 (SE=9.28) (Figure 3) and 95% fixed kernel home ranges averaged 80 km^2 (SE=18.75) (Figure 4).

Table 1. Summary information from mountain goat collaring operation, Kodiak Island, Alaska.

ID	Collar type	Capture date	Sex	Capture lat	Capture long	Capture site	Age
IG16	Telonics SST	7/20/2015	F	57.83018	-152.54569	Sheratin	9
IG17	Telonics SST	7/20/2015	M	57.75428	-152.75899	Powerline pass	6
IG18	ATS GlobalStar	7/20/2015	F	57.68758	-152.74717	Sargent Creek	9
IG19	Telonics SST	7/21/2015	F	57.55519	-152.38627	Marin Range	10
IG20	ATS GlobalStar	7/21/2015	M	57.53786	-152.59532	Kalsin Ridge	4
IG21	ATS GlobalStar	7/21/2015	F	57.62791	-152.92125	S. of Crown Mt.	3
IG22	ATS GlobalStar	7/21/2015	M	57.57773	-152.97446	E. of Widgeon Bowl	1
IG23	ATS GlobalStar	7/21/2015	M	57.3742	-153.10243	North arm of Kiliuda Bay	6
IG24	ATS GlobalStar	7/21/2015	F	57.36534	-153.18416	North arm of Kiliuda Bay	2
IG25	ATS GlobalStar	7/21/2015	F	57.4771	-153.24891	Uganik Ridge	9
IG26	ATS GlobalStar	7/21/2015	F	57.47355	-153.23971	Upper Uganik	7
IG27	ATS GlobalStar	7/24/2015	F	57.64995	-153.13991	S. of Terror Lake	8
IG28	ATS GlobalStar	7/24/2015	M	57.5916	-153.19109	Uganik/Terror Lake	4
IG29	ATS GlobalStar	7/24/2015	F	57.46939	-153.20273	Uganik	1
IG30	ATS GlobalStar	7/24/2015	F	57.0168	-153.91808	Hepburn	4
IG31	Telonics SST	7/24/2015	M	56.91988	-153.7107	Aliulik Pen.	6
IG32	ATS GlobalStar	7/24/2015	M	56.94525	-153.62815	Aliulik Pen.	2
IG33	ATS GlobalStar	7/24/2015	F	57.09977	-153.74099	E. of Deadman Bay	4
IG34	ATS GlobalStar	7/24/2015	F	57.16579	-153.84981	W. of Deadman Bay	11
IG35	ATS GlobalStar	7/24/2015	F	57.15709	-153.73825	Upper Deadman Bay	6
IG36	Telonics SST	7/25/2015	F	57.18069	-154.08055	S. of Frazer Lake	3
IG37	ATS GlobalStar	7/25/2015	F	57.24579	-154.23909	Frazer/Red Lake	3
IG38	ATS GlobalStar	7/25/2015	F	57.38071	-153.68616	E. of Uyak Bay	12
IG39	ATS GlobalStar	7/25/2015	M	57.29161	-154.09599	S. Karluk	7
IG40	Telonics SST	7/25/2015	M	57.37227	-153.64966	E. of Uyak Bay	7
IG41	ATS GlobalStar	7/25/2015	F	57.36792	-153.58956	E. of Uyak Bay	9
IG42	ATS GlobalStar	7/25/2015	F	57.39191	-153.40027	N. of Windy Lake	6
IG43	ATS GlobalStar	7/26/2015	F	57.58195	-153.64374	Zachar Pen.	6
IG44	ATS GlobalStar	7/26/2015	M	57.44758	-153.64322	Brown's Ridge	6
IG45	ATS GlobalStar	7/26/2015	M	57.37695	-153.43768	Upper Zachar	10
IG46	ATS GlobalStar	7/26/2015	M	57.30633	-153.46106	Upper Zachar	5

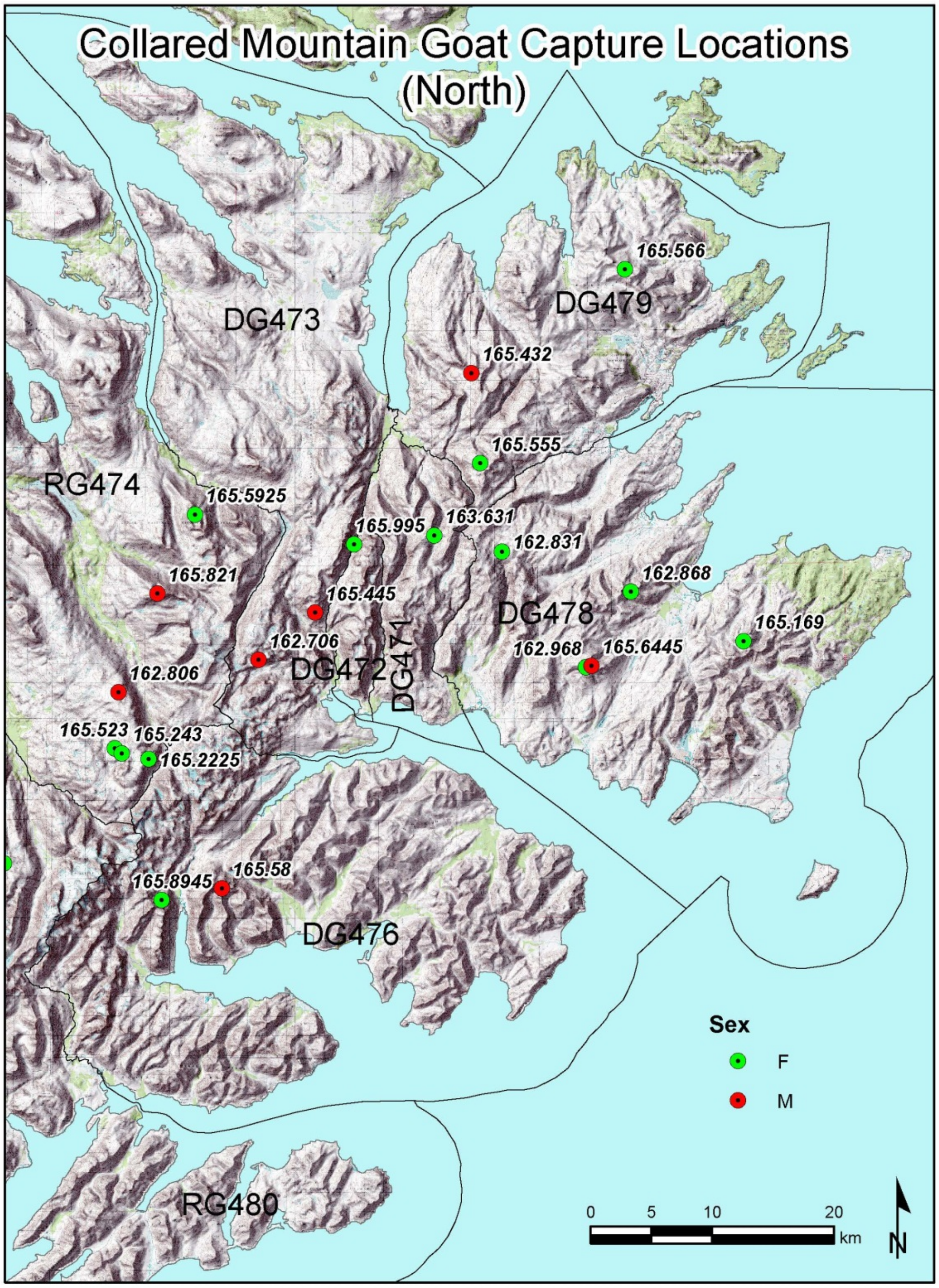

Figure 1. Locations and radio frequencies of female (yellow) and male (red) mountain goats captured on the northern portion of Kodiak Island, Alaska, July 2015.

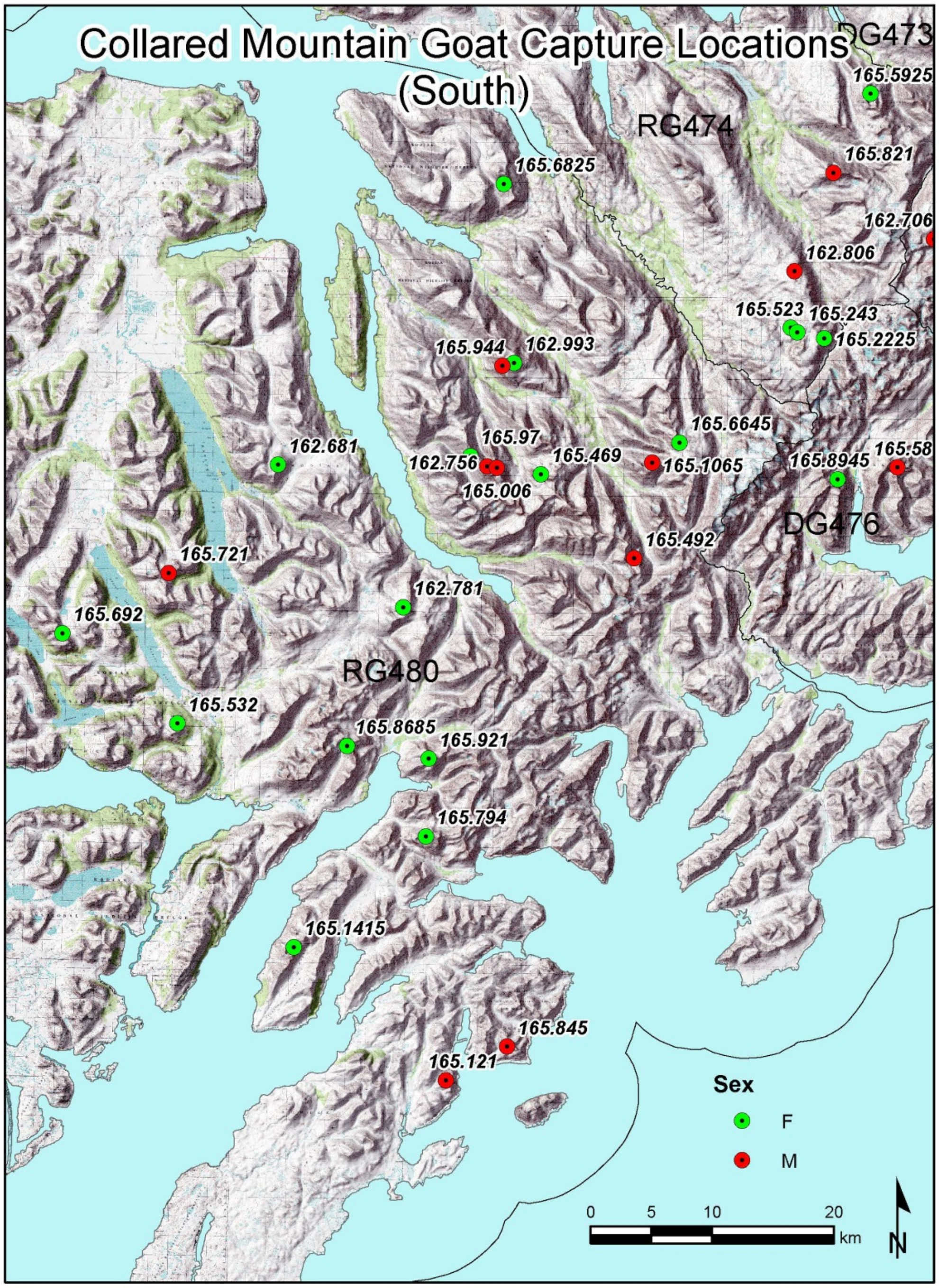

Figure 2. Locations and radio frequencies of female (yellow) and male (red) mountain goats captured on the southern portion of Kodiak Island, Alaska, July 2015.

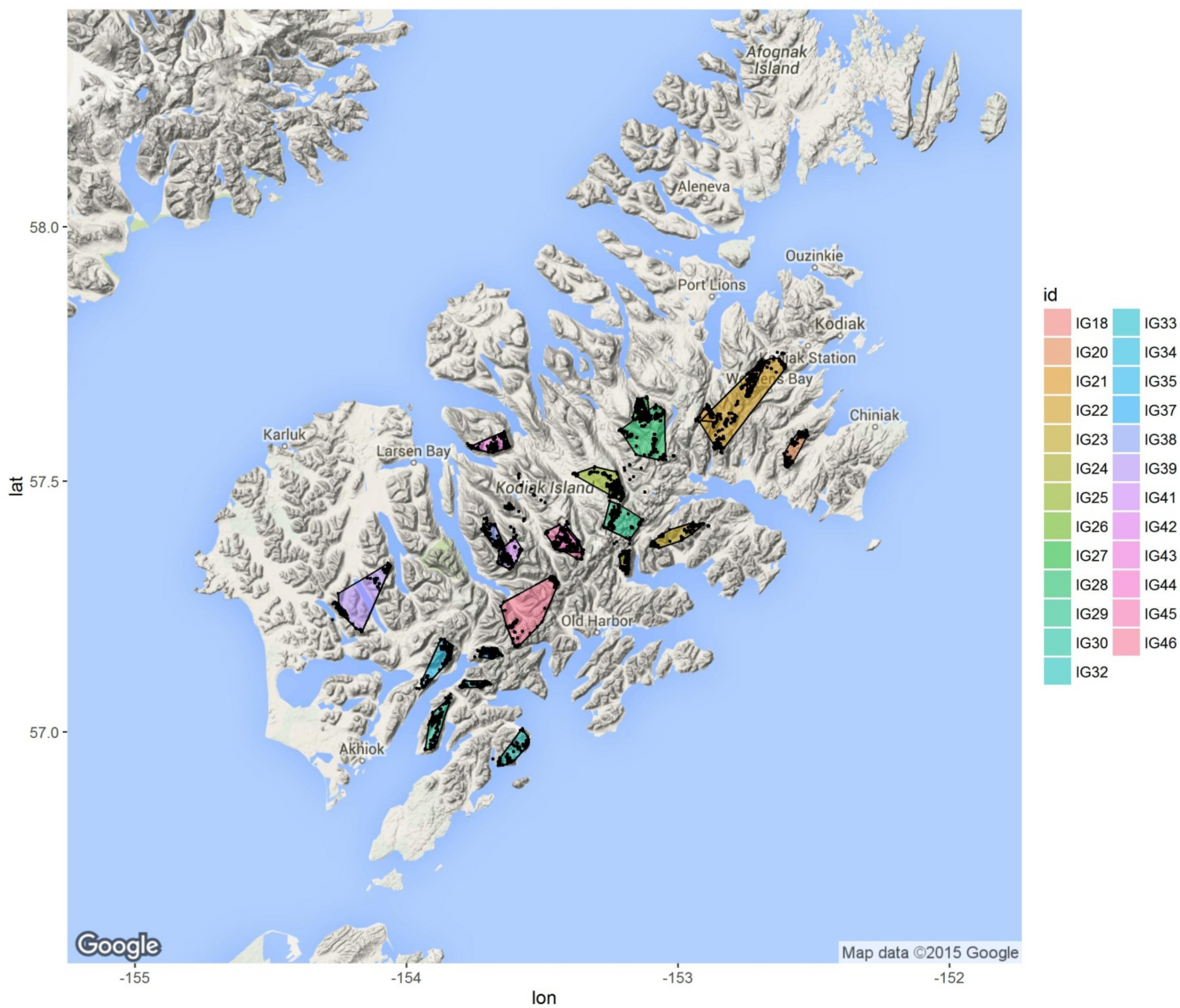

Figure 3. Minimum convex polygon (95%) home ranges of Globalstar GPS-collared mountain goats (n=25), Kodiak, Alaska, July-December 2015.

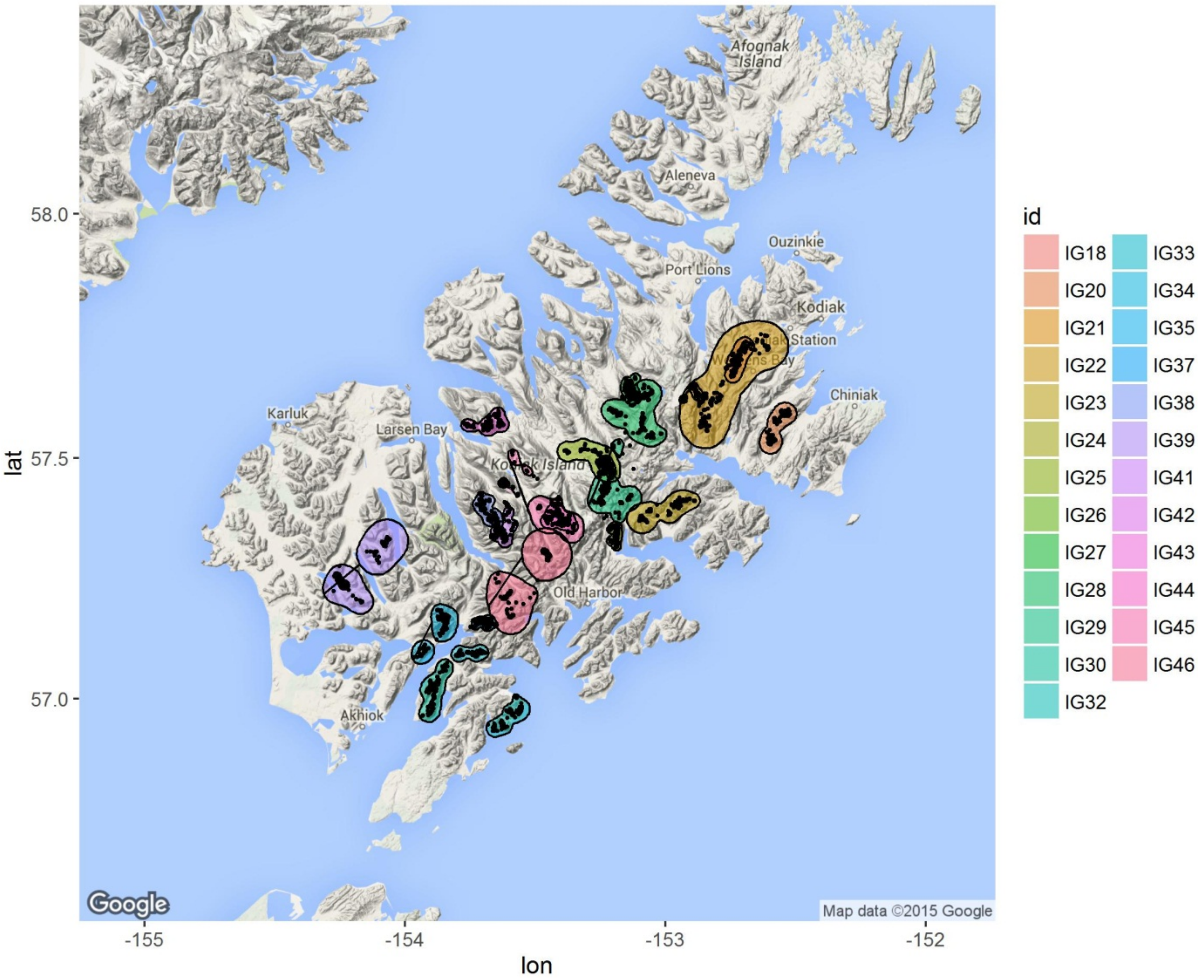

Figure 4. Fixed kernel (95%) home ranges of Globalstar GPS-collared mountain goats (n=25), Kodiak, Alaska, July-December 2015.